



Digital Diary: Building Your Own Contact Book

Python Project – Contact Book App



The Lost Number Problem

Imagine you forgot your friend Aarav's phone number and need to call him for homework help!
How do we keep all important numbers safe and easy to find?

What is a Digital Diary?

A Digital Diary is like a modern address book on your computer! It stores information as Keys and Values.

- Kirana store ledger: Customer name → Amount owed
- Aadhaar card: Unique number → Your details
- Mobile contacts: 'Mummy', 'Daddy' → Phone numbers

Key = Label, Value = Information stored!





Python Dictionaries Refresher

A dictionary stores data as pairs — Keys and Values!

Example:

```
contacts = {'Daddy': '9876543210', 'Mummy':  
            '9123456789', 'Diya': '9988776655'}
```

You already know how to use these!



Let's Practice!

Here are some contacts:

- Aarav: 9876543210
- Saanvi: 9123456789
- Arjun: 9988776655

Can you tell the phone number for Saanvi?

Starting Our Contact Book

Let's begin with an empty dictionary:

```
contacts = {}
```

Our Menu Options:

1. Add Contact
2. View All Contacts
3. Search Contact
4. Exit

We will build this step by step!



Adding Contacts

Here's how we add a new contact to our dictionary:

```
def add_contact():  
    name = input("Enter name: ")  
    phone = input("Enter phone number: ")  
    contacts[name] = phone  
    print("Shabaash! Contact added.")
```

The name becomes the key, and the phone number becomes the value!



Viewing All Contacts

```
def view_contacts():  
    for name, phone in contacts.items():  
        print(f"{name}: {phone}")
```

The 'for loop' goes through each contact one by one and prints the name with its phone number!





Our Menu Loop

Think of a while loop as a door that stays open until you decide to leave! Our menu keeps showing options until you choose Exit.

```
while True:
    print("1. Add 2. View 3. Search 4. Exit")
    choice = input("Pick: ")
    if choice == "4":
        break
```

Searching Contacts – The Truecaller Logic

Just like Truecaller helps you find who's calling, our search feature finds contacts instantly!

```
def search_contact():  
    name = input("Enter name: ")  
    if name in contacts:  
        print(f"Number: {contacts[name]}")  
    else:  
        print("Contact not found.")
```



Recap: What We Learned Today

- Use dictionaries to store contacts
 - Keys = Names, Values = Phone numbers
 - Add, View, Search functions built
 - UI loop keeps app running until exit
- Shabaash! You built your first digital diary!





An illustration of five diverse children (three boys and two girls) smiling and using laptops. They are positioned on the left side of the image. A large, white, cloud-like speech bubble with a black outline is on the right, containing the text 'Your Turn to Build!'. The background is a warm orange with stylized white clouds and sparkles.

Your Turn to Build!

Now it's your turn! Start coding your contact book and make it yours. Be creative, explore new ideas, and don't be afraid to try different things. You have all the skills you need!

Shabaash, Young Coders!

Thank You!

Any questions? See you in
Session 2 for more coding fun!